ofmphy: An Open-Source Simulation of an 800G-Class Coherent Optical Transceiver

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***Abstract*—Interoperable 800G coherent standards (OIF 800ZR, OpenZR+, Open ROADM) share a single line format: dual-polarisation 16QAM near 118 GBd built from a common set of transmitter, forward-error-correction, channel and receiver-DSP blocks. The specifications are public, yet no open-source implementation of the complete physical layer has been available. This paper presents *ofmphy*, a pure-numpy, fully vectorised baseband simulation of such a transceiver. The transmitter implements Gray-mapped DP-16QAM, root-raised-cosine shaping and a bit-exact encoder for the standard open FEC (oFEC). The channel carries chromatic dispersion, polarisation rotation with differential group delay and polarisation-dependent loss, laser phase noise, amplified spontaneous emission, transmitter and receiver I/Q imperfections, an ADC model and an optional split-step Manakov fibre. The blind receiver chains front-end correction, dispersion compensation, a widely-linear CMA/RDE/DD-LMS butterfly equalizer, blind phase search, reference-aided alignment, per-polarisation demapping and soft- or hard-decision oFEC decoding. The measured soft-decision waterfall maps pre-FEC bit error rates from 2.3×10^{-2} to 1.95×10^{-2} onto output rates from 10^{-2} to 1.1×10^{-6}, within about 0.1 dB of the standard's qualified 2.0×10^{-2} point, and the end-to-end chain reaches that point 0.61 dB from the additive-noise bound, inside the 0.4–0.8 dB band reported for idealised-front-end simulations. The package is small enough to read and fast enough to sweep, and it is released for others to complete.**

***Index Terms*—Coherent optical communications, digital signal processing, forward error correction, 800ZR, open-source software, simulation.**

I. Introduction

THE 400ZR generation of coherent interfaces made interoperability a design goal rather than an afterthought, and the 800G generation that followed (OIF 800ZR [1], OpenZR+ [2], Open ROADM [3]) kept that direction: a shared line format of dual-polarisation 16QAM near 118 GBd, a common transmitter and receiver-DSP architecture, and a single standardised soft-decision FEC, the open FEC (oFEC), carried in ITU-T G.709.x [4]. The specifications are public and defined to the bit, yet the complete physical layer has had no open implementation. Decoder studies report FEC thresholds and floors [5], [6], and network-level planners are open [7], but a researcher who wants to reproduce the standardised line end to end, from encoder to demapper, has had to work from the specification PDFs.

ofmphy provides that baseband physical layer as readable, vectorised code. It is deliberately small, about three thousand lines of numpy across a dozen modules, each block written in its textbook form and vectorised so that a frame of 178 000 information bits runs through transmitter, channel, receiver and both decoders in roughly two seconds on one core. The blocks are treated uniformly: the oFEC encoder and decoder are two blocks among the mapper, the pulse shaper, the equalizer, the carrier-recovery and demapping stages, each with its own equations and its own unit test. The contributions are:

- a complete, inspectable transmitter–channel–receiver chain for the interoperable 800G line format, in pure numpy;
- a bit-exact open implementation of the standard oFEC, its encoder address equations, serialisation rectangles and startup rule, with soft Chase–Pyndiah and hard bounded-distance decoders;
- a blind receiver with widely-linear equalization for transmit-side I/Q imperfections, specified block by block; and
- a validation methodology anchored on three sides: textbook theory, the standards' numbers, and published hardware and simulation results.

The library is meant to grow by contribution: Section VII lists the blocks it does not yet have, in the hope that readers adopt them.

II. Related Work

OptiCommPy [8] covers general fibre-optic communication experiments in Python, and GNPy [7] models amplified networks at the quality-of-transmission level; neither implements a standardised FEC. oFEC performance itself has been studied in simulation [5], [6], [4], and commercial encoder/decoder IP exists, but the author is not aware of a prior open implementation of the full standardised chain, encoder equations and receiver DSP together. Commercial simulators cover the chain but are neither free nor inspectable. ofmphy sits between these: not a network planner, not a laboratory replacement, but a readable executable form of the standard line interface.

III. Transmitter and Channel

Fig. 1 shows the chain. Payload bits are oFEC encoded, then scrambled by a fixed pattern: the FEC frame has structure (parity columns, an all-zero startup region), and every blind algorithm downstream assumes the symbol stream looks random. The scrambled bits are mapped onto DP-16QAM at $R_s = 118.203350603$ GBd with Gray labelling, so that neighbouring constellation points differ in a single bit and a symbol error usually costs one bit. Root-raised-cosine (RRC) shaping with 5% roll-off then confines the spectrum to barely more than the symbol rate, which is what lets these channels pack a dense WDM grid and why two samples per symbol are enough throughout the model. The frame is processed circularly, which removes filter transients and makes dispersion exactly invertible; the laser phase walk is pinned into a

Brownian bridge and the frequency offset snapped to the FFT grid so the circular frame has no seam.

A. oFEC encoding

The FEC is the standard oFEC, a braided product-like convolutional code over an extended BCH(256,239) component. Bits occupy a semi-infinite matrix of 128 columns viewed as 16×16 square blocks; the codeword at row pair (R, r) rereads 128 "front" bits from earlier rows and writes 111 fresh information bits and 17 parity bits into row R, following the address maps

$$W_{R,r}(k) = V\big((R\oplus 1) - 20 + 2\lfloor k/16\rfloor,\ \lfloor k/16\rfloor, (k \bmod 16)\oplus r, r\big),\ \ k = 0,\dots,127, \tag{1}$$

$$W_{R,r}(128{+}j) = V\big(R, \lfloor j/16\rfloor, r, (j \bmod 16)\oplus r\big),\ \ j = 0,\dots,127, \tag{2}$$

where $\oplus$ is bitwise XOR and $V(\cdot)$ addresses {block row, block column, row, column}. Every bit is protected by exactly two codewords 5 to 21 block rows apart, giving rate $111/128 = 0.867$, and the $\oplus r$ twist rotates bit positions between rows so that a burst in one row lands at different positions of different component words. This braiding is what makes iterative decoding effective: correcting one codeword sharpens half the bits of words up to 21 block rows away, so reliability flows along the stream in both directions as the sweeps repeat. The nearest front row sits five block rows back, a guard that gives hardware room to pipeline; the same separation is what lets this package decode whole batches of codewords at once. The component code extends BCH(255,239), generator $g(y) = y^{16}+y^{14}+y^{13}+y^{11}+y^{10}+y^{9}+y^{8}+y^{6}+y^{5}+y+1$ over $\mathrm{GF}(2^8)/0\mathrm{x}11\mathrm{d}$, by an overall parity bit. The extension raises $d_{\min}$ from 5 to 6: the code still corrects $t = 2$ errors, but far fewer triple-error patterns miscorrect, which matters when Chase decoding later feeds the algebraic decoder deliberately flipped words. The first 20 rows of a stream encode against an all-zero front, the startup rule $WH' = 0$, so that both ends assume the same fictitious history and a receiver can join at the head of a stream. Fig. 2 shows the geometry. The implementation carries the specification's input and output serialisation rectangles, so the transmitted bit order is the standard's, not a convenient permutation.

B. Fibre and noise

In the fibre, group velocity depends on frequency, so each pulse spreads over its neighbours; in baseband this chromatic dispersion is the quadratic all-pass phase $H_{\mathrm{CD}}(\omega) = \exp\!\Big(+j\frac{\beta_2}{2}L\omega^2\Big)$ with $\beta_2 = -D\lambda^2/(2\pi c) = -21.7\ \mathrm{ps}^2/\mathrm{km}$. The polarisation channel is built as rotation, delay, rotation: the unitary rotations set an arbitrary state of polarisation, a differential group-delay element, first-order polarisation-mode dispersion, delays one principal state against the other, and 1 dB of polarisation-dependent loss attenuates one axis, which breaks unitarity and leaves the two tributaries at different SNR at the receiver. Transmitter and local-oscillator phase noise act as one Wiener walk of twice the 100 kHz per-laser linewidth, since the two independent walks add, with a 1 GHz carrier offset between the free-running lasers; amplified spontaneous emission is white noise calibrated through the optical signal-to-noise ratio (OSNR) in its industry reference bandwidth of 0.1 nm, 12.5 GHz at 1550 nm, via $E_s/N_0 = \mathrm{OSNR}\cdot B_{\mathrm{ref}}/R_s$. Setting `nl_steps` replaces the linear fibre with a symmetric split-step integration of the Manakov equation,

$$\frac{\partial A}{\partial z} = -\frac{\alpha}{2}A - j\frac{\beta_2}{2}\frac{\partial^2 A}{\partial t^2} + j\frac{8}{9}\gamma\left(|A_x|^2 + |A_y|^2\right)A, \tag{3}$$

using lumped per-span EDFAs; the $8/9$ factor is the Kerr coefficient averaged over the fast polarisation scrambling of real fibre. The budget $\mathrm{OSNR} = P_{\mathrm{launch}} - \mathrm{loss} + 58 - \mathrm{NF} - 10\log_{10} n_{\mathrm{spans}}$ ties ASE to launch power, so raising the power buys OSNR linearly while the Kerr distortion grows with its cube, which is what creates the optimum of Section V.

C. Electrical front ends

Both electrical front ends are imperfect, because modulator arms, 90° hybrids and photodiode pairs never match exactly; the model is the same per polarisation at transmitter and receiver. The Q rail carries a gain and quadrature-angle error,

$$y = I + jge^{j\phi_q}Q \;=\; \alpha x + \beta x^*, \tag{4}$$

so the conjugate term is a mirrored copy of the signal spectrum sitting $20\log_{10}|\beta/\alpha|$ dB below it. The rails are also differentially delayed by a skew τ_s, whose image term $j\sin(\omega\tau_s/2)\,X^*(-\omega)$ grows toward the band edge, so at 118 GBd even 1 ps of skew matters at the edges of the band. Defaults are 0.5 dB, 2° and 1 ps per side, with a 6-bit ADC that clips at 3.2σ, near where clipping distortion and quantisation noise balance for a signal with Gaussian-like tails.

IV. Receiver DSP

The receiver is blind except where the simulation substitutes its known reference sequence for the pilot machinery of a real framer, namely frame alignment, polarisation pairing, quadrant rotation and noise-variance calibration. The nine blocks below run in order, and the order is deliberate: each stage removes the impairment that would break the assumptions of the next, so the bulk static effects (skew, dispersion) go first, blind statistical corrections next, and the decision-based stages last, once the constellation is clean enough to slice. Each block is written in its textbook form and vectorised over blocks of symbols rather than looped per symbol.

A. Front-end correction

The receive I and Q rails arrive slightly misaligned in time and not perfectly orthogonal. The block first removes the differential rail delay using the skew value the module measures at manufacture, then restores orthogonality blind by Gram–Schmidt [9]:

$$I' = I/\sqrt{\mathbb{E}[I^2]}, \qquad Q'' = Q - \mathbb{E}[I'Q]\,I', \tag{5}$$

$$Q' = Q''/\sqrt{\mathbb{E}[Q''^2]}. \tag{6}$$

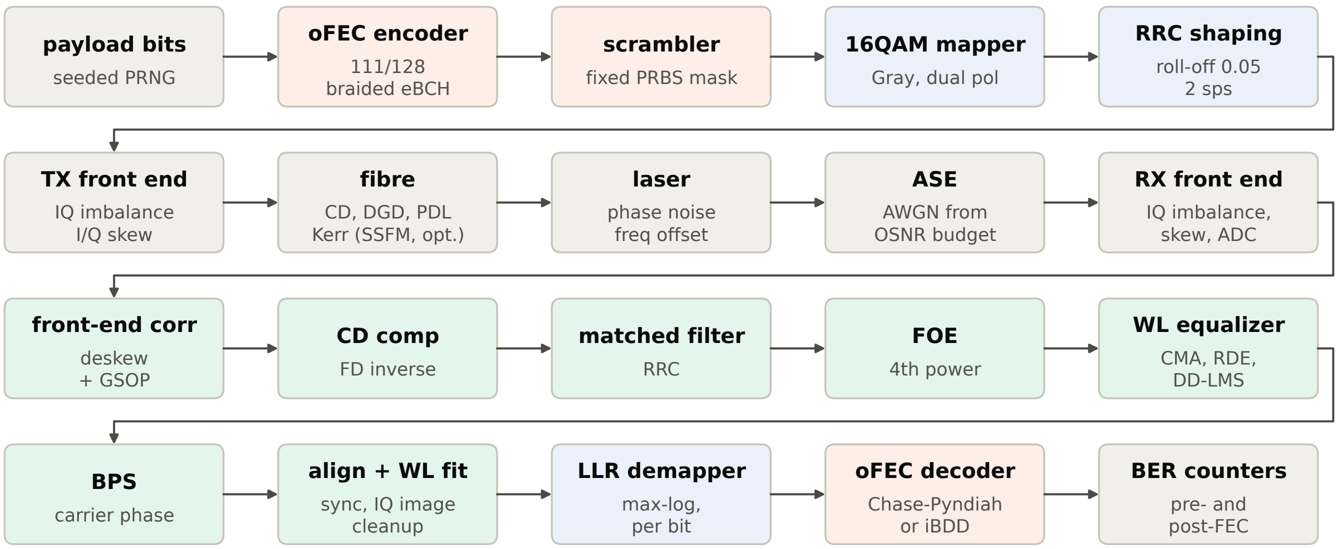

Fig. 1. System overview: transmit path, channel with both electrical front ends, and the blind receiver with its decoders. The oFEC encoder and decoder are two blocks among the rest.

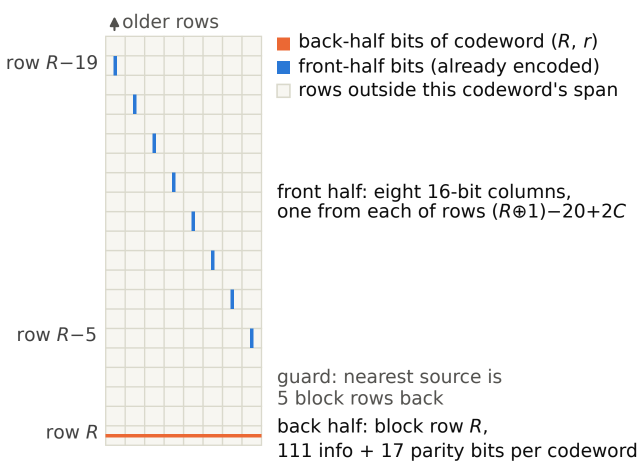

Fig. 2. The oFEC matrix with one codeword picked out: its back half in its own block row, its front half read as one 16-bit column from each of eight earlier rows.

Afterward the rails are uncorrelated and equal in power. Because these are second-order statistics, the correction holds even while a frequency offset spins the constellation, so the block can run first, ahead of carrier recovery. The one thing it cannot see is the transmitter's own I/Q image, which is left to the equalizer (block E) and the post filter (block G).

B. Chromatic dispersion compensation

Chromatic dispersion is a known, static, all-pass phase, so it is removed exactly and once in the frequency domain,

$$H_{\mathrm{CDC}}(\omega) = \exp\!\left(-j\,\tfrac{\beta_2}{2} L_{\mathrm{tot}}\,\omega^2\right), \tag{7}$$

rather than left to the short adaptive equalizer. At 80 km its impulse response spans about 150 symbols, far longer than the 21-tap equalizer could absorb.

C. Matched filtering

The receive-side root-raised-cosine filter. Cascaded with the identical transmit filter it forms a raised cosine, which is Nyquist (zero intersymbol interference at the sampling instants) and at the same time maximises signal-to-noise ratio against the white ASE.

D. Frequency offset estimation

The transmitter and local oscillator are not frequency-locked. Raising 16QAM to the fourth power cancels its $\pi/2$ rotational symmetry and leaves a discrete tone at four times the residual offset,

$$\Delta f = \tfrac{1}{4}\,\arg\max_f \left| \textstyle\sum_{\mathrm{pol}} \mathcal{F}\{y(t)^4\} \right|, \tag{8}$$

located by an FFT peak search with parabolic interpolation to sub-MHz accuracy. For 16QAM the fourth power does not cancel the modulation exactly, as it does for QPSK, because the symbols sit on three rings; enough of the tone survives to stand clear of the modulation noise around it. The small remaining error is absorbed by the phase search downstream.

E. Widely-linear butterfly equalizer

A single adaptive stage inverts the polarisation channel, the residual dispersion and the transmitter I/Q image together. It is a 2×2 butterfly with 21 taps per branch, spaced at $T/2$ so that its response does not depend on where within the symbol the ADC happens to sample, but widely linear: each output filters both inputs and their complex conjugates,

$$\begin{aligned} y_i[n] = \sum_{p\in\{x,y\}} \sum_t \Big(& w_{ip}[t]\, x_p[2n{+}t] \\ & + v_{ip}[t]\, x_p^*[2n{+}t] \Big), \end{aligned} \tag{9}$$

because transmitter I/Q imbalance and skew reach the receiver as a conjugate mixing $\alpha x + \beta x^*$ that a strictly linear equalizer cannot represent. Adaptation is a three-stage cascade of block-averaged gradients. Constant-modulus acquisition and then radius-directed refinement drive the outputs onto the 16QAM moduli,

$$e_i^{\mathrm{CMA}} = R_2 - |y_i|^2, \quad R_2 = \mathbb{E}|s|^4/\mathbb{E}|s|^2, \tag{10}$$
$$e_i^{\mathrm{RDE}} = r_{\mathrm{ring}}^2 - |y_i|^2, \tag{11}$$

both blind to carrier phase, so they converge ahead of carrier recovery. CMA treats the constellation as one average ring, which keeps it stable while the eye is still closed; RDE takes over once the output is clean enough to assign each symbol to one of the three 16QAM radii. The conjugate branch is held frozen through acquisition, where widely-linear CMA can otherwise lock onto the image, and released for the radius-directed stage. Once blind phase search provides a carrier track $\phi[n]$, a decision-directed LMS pass tightens the taps against sliced decisions in the derotated domain; unlike the modulus criteria it sees the full error vector, so it picks up the residual ISI they cannot,

$$z = y\, e^{-j\phi}, \quad e = \mathrm{dec}(z) - z, \tag{12}$$
$$\mathbf{W} \leftarrow \mathbf{W} + \mu\, \langle e\, e^{+j\phi} \mathbf{x}^* \rangle_{\mathrm{block}}. \tag{13}$$

A singularity check rebuilds one output as the orthogonal complement of the other if constant-modulus adaptation collapses both onto the same polarisation.

F. Blind phase search

Carrier phase is recovered blind and jointly across the two polarisations, which share one local oscillator and therefore one phase, so the two distance metrics are summed and the estimate steadies. Only a quarter turn needs searching, since a square QAM constellation repeats every $\pi/2$; the absolute quadrant is fixed later by alignment. The block tests 32 phases ϕ_b across that quarter turn and, for each symbol, keeps the phase minimising the windowed distance to the nearest constellation points,

$$\begin{aligned} D[n,b] = \sum_{\mathrm{pol}} \sum_{|k| \le W/2} \Big| \, & y[n{+}k]\, e^{-j\phi_b} \\ & - \mathrm{dec}\big(y[n{+}k]\, e^{-j\phi_b}\big) \Big|^2, \end{aligned} \tag{14}$$

summed over a 96-symbol circular window and both polarisations, with a $\pi/2$ unwrap of the resulting track. The window length trades noise averaging against tracking bandwidth: longer windows smooth the decision noise but lag the laser walk, and 96 symbols suits the 200 kHz combined walk at this symbol rate. A decision-aided maximum-likelihood refinement [10] is implemented but measured to add nothing at the 100 kHz linewidth modelled here; it stays available for MHz-class lasers.

G. Alignment and widely-linear post filter

A blind receiver recovers the signal only up to a polarisation swap, a bulk delay, a quadrant rotation and a complex gain. These are resolved by circular cross-correlation against the known transmit sequence,

$$c_{ij}[k] = \mathcal{F}^{-1}\big\{\mathcal{F}\{y_i\}\, \mathcal{F}\{\mathrm{ref}_j\}^*\big\}, \tag{15}$$

the simulation's stand-in for the pilot machinery of a real framer. A short 5-tap widely-linear least-squares filter, fitted per polarisation on 2048-symbol blocks, then cleans up whatever transmitter I/Q image survives to the symbol rate,

$$\hat{s}[n] = \sum_t \big(p_t\, z[n{+}t] + q_t\, z^*[n{+}t]\big), \tag{16}$$

the batch equivalent of a widely-linear decision-directed loop. The fit is per block rather than global because a conjugate term does not stay put: it counter-rotates against the carrier, so after derotation it drifts at twice the residual phase, slowly enough for each block to track.

H. Demapping

The demapper produces one soft bit per coded bit for the FEC. Gray 16QAM factors into two independent 4-level amplitude (PAM4) rails, one on I and one on Q, so the max-log rule is applied per rail, with the noise variance estimated from the reference error vector separately for each polarisation, since PDL leaves the two at different SNR,

$$L(b) = \frac{\min_{s \in \mathcal{S}_b^1} (u{-}s)^2 - \min_{s \in \mathcal{S}_b^0} (u{-}s)^2}{2\sigma_d^2}. \tag{17}$$

The max-log approximation is information-lossless for the Gray labelling used here [11], so nothing is gained by the exact log-sum-exp.

I. FEC decoding

The standard fixes only the encoder; two decoders are provided, both vectorised over 80-codeword batches, since codewords fewer than five block rows apart share no bit. The soft decoder is Chase–Pyndiah turbo product decoding [12]. An algebraic decoder on its own cannot use soft information, so each component word is decoded 2^6 times, once per pattern of flips on its six least-reliable bits, which turns syndrome-table bounded-distance decoding into a search over the codewords most likely to have been sent. Candidates are scored by analog weight, and the updated reliability of a bit is the score gap between the winning codeword and the best candidate that disagrees with it,

$$m(c) = \sum_{i:\, c_i \neq h_i} |\Lambda_i|, \tag{18}$$
$$\Lambda_j' = \big[\min_{c:\, c_j \neq d_j} m(c) - m(d) \big]\, \mathrm{sgn}_j, \tag{19}$$

exchanged over four soft-in soft-out sweeps between the two roles of each bit, with a schedule that trusts the extrinsic values more, and falls back on a default reliability less, as the sweeps converge; two hard clean-up sweeps finish, the shape of the G.709.3 reference decoder [4]. The hard decoder is plain iterated bounded-distance decoding, an order of magnitude faster at about 1.5 dB cost in threshold (Fig. 3). Finite frames add a fixed 20-row leader (absorbing the startup transient) and 21 termination rows (the coupling span), counted as line bits but not payload.

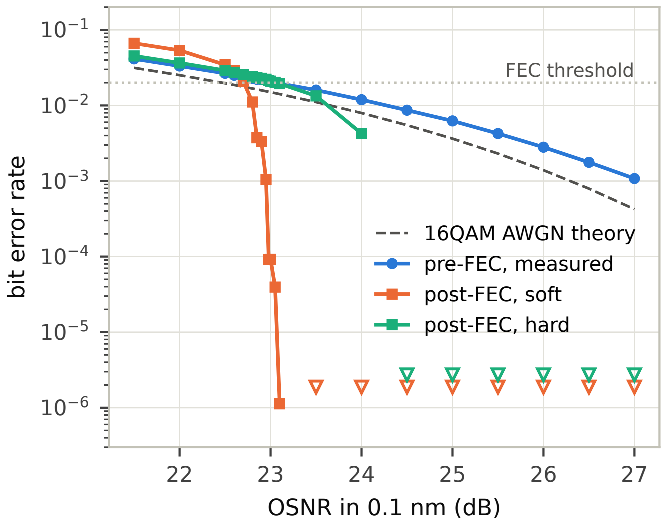

Fig. 3. Measured OSNR waterfall. The soft-decision output descends through four decades as the input crosses the oFEC threshold (dotted); open markers are counting floors where no errors were observed.

TABLE I
DISTANCE TO THE STANDARD AT PRE-FEC 2.0×10^{-2} (REGENERATED BY `docs/standards_gap.py`).

Chain variant	OSNR req.	Distance
Ideal AWGN 16QAM (theory)	22.47 dB	—
Blind DSP only	22.62 dB	+0.16 dB
+ fibre, PMD, laser	22.71 dB	+0.26 dB
Full chain, compensated	23.08 dB	+0.61 dB
800ZR compliance limit	27.0 dB	3.92 dB margin

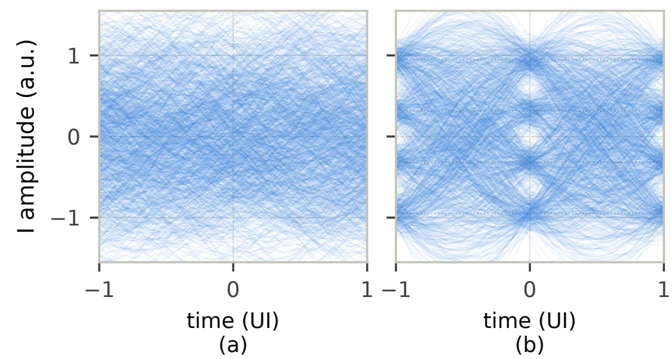

Fig. 4. Eye diagrams of one I rail at 8 samples/UI (OSNR 26 dB): (a) closed before the equalizer; (b) reopened into the four PAM4 levels after the full DSP, with the ideal levels dotted. With 5% roll-off the eye is open only near the sampling instant $t = 0$.

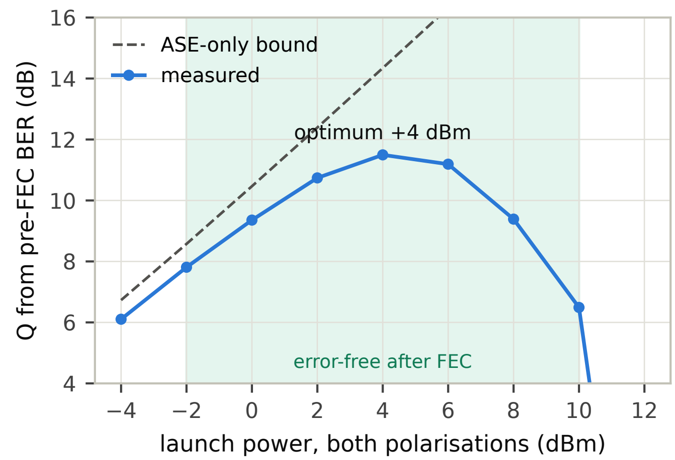

Fig. 5. Q versus launch power over 10×80 km with the split-step Manakov fibre. The dashed line is the ASE-only bound; the shaded band is the region the FEC returns error-free.

V. VALIDATION AND RESULTS

Validation is anchored three ways. First, with every impairment disabled, the measured pre-FEC bit error rate (BER) reproduces the textbook Gray-16QAM curve within counting noise; this is a unit test, not a figure. Second, the standards: Fig. 3 shows the OSNR sweep with every impairment on. Transition points carry up to 17.8 million payload bits, so the soft curve is measured, not extrapolated, through the low decades: output BER 1.1×10^{-2} at input 2.27×10^{-2}, 1.1×10^{-3} at 2.14×10^{-2}, 9×10^{-5} at 2.06×10^{-2}, and 1.1×10^{-6} at 1.95×10^{-2}, with no errors observed beyond. A 0.3×10^{-2} slice of input maps onto four output decades, within about 0.1 dB of the reference decoder's qualified 2.0×10^{-2} point [2]; published spreads between decoder variants of this code are 0.15–0.9 dB [5], [6], and published stall-pattern floors (10^{-9}–10^{-13} without post-processing [6]) are the reason the 10^{-15} contract is quoted from the standard rather than claimed from simulation.

Table I reads the same data against the operating point. The full chain needs 23.08 dB OSNR to reach pre-FEC 2.0×10^{-2} against a 22.47 dB theory bound: a 0.61 dB implementation penalty, decomposed by switching subsystems off. The blind DSP costs 0.16 dB, propagation impairments almost nothing once compensated, and the front-end stack (dominated by the 6-bit ADC floor and the transmit-skew residual) the remaining 0.4 dB. This sits inside the 0.4–0.8 dB band reported for idealised-front-end simulation studies [13], while real 118–120 GBd hardware measures 24–26 dB back-to-back, 25 dB for the cleanest published component demonstration [14]; the missing 1.5–2.5 dB lives in transmitter bandwidth, linearity and converter effects that are out of scope here. The margin to the 27 dB 800ZR receiver compliance limit is 3.9 dB, the expected sign and size for a behavioural model.

Third, the physics on either side of the decoder behaves. The eye diagrams of Fig. 4 close completely before the equalizer and reopen into the four PAM4 levels at the sampling instant after the full DSP. With the split-step fibre enabled over 10×80 km and OSNR tied to launch power, Q against launch power traces the classic bell (Fig. 5): optimum +4 dBm at $Q = 11.3$ dB, error-free payloads from -2 to $+10$ dBm, collapse by +12 dBm. The optimum lands within 0.6 dB of a closed-form GN-model evaluation [15] of the same link, and published DP-16QAM measurements over 80 km spans scale to +4.1 to +5.0 dBm per 118 GHz of occupied spectrum [16], bracketing it.

VI. IMPLEMENTATION NOTES

Everything is numpy; there is no compiled extension. Vectorisation follows two patterns. Structure-parallel: the oFEC decoders batch all codewords of five consecutive block rows (proved bit-disjoint) times all Chase patterns into single array operations over Galois-field lookup tables. Block-parallel: adaptive loops (CMA, RDE, DD-LMS) update on block-averaged gradients, one numpy call per 64-symbol block. Averaging the gradient leaves the fixed point of the adaptation unchanged and only narrows its tracking bandwidth, which a channel static over one frame tolerates, so the blocked loop lands on the per-symbol steady state at a fraction of the interpreter cost. A 100-row frame (177 600 payload bits) runs the full chain with both decoders in about two seconds on a single laptop-class core. The repository ships 49 unit tests, including bit-level checks of the interleaver equations against the specification's properties (tiling, double coverage, coupling span) and an end-to-end theory-match test; every figure and table in this paper regenerates from committed scripts with fixed seeds.

VII. Limitations and Roadmap

The simulation is symbol-synchronous: clock offset and timing recovery (a fractional resampler with a Gardner or Godard loop) are the most valuable missing blocks. Also absent: DSP framing and pilots (the reference-aided alignment stands in for them), probabilistic constellation shaping for OpenZR+ higher modes, laser RIN, DAC quantisation, transmitter bandwidth and nonlinearity models (the largest hardware-versus-simulation gap), WDM neighbours, and official interoperability test vectors for the oFEC serialisation, which the author would particularly welcome help verifying against hardware. The code is structured so that each of these is one module or one config knob away.

VIII. Conclusion

ofmphy makes the interoperable 800G physical layer executable: a complete transmitter, channel and blind receiver, with the standard oFEC as one of its blocks, and measurements that line up with theory on one side and the standards and published results on the other. It is offered as a base that anyone can verify, break and extend.

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